

# DAY 1 — PYTHON FUNDAMENTALS

Variables • Data Types • input() • print() • Operators

**MODULE** Python

**PRACTICE** 20 questions

**TASK** Build Calculator

## What You'll Learn Today

- ✓ How to create and use variables to store data
- ✓ Understanding different data types (int, float, str, bool)
- ✓ Taking input from users with input()
- ✓ Displaying output with print()
- ✓ Working with arithmetic, comparison, and logical operators
- ✓ Building a practical calculator application

# 1 VARIABLES — Containers for Data

## What is a Variable?

A variable is like a labeled box that stores a value in memory. You can store data in it, use it, and change it anytime.

## Syntax

```
variable_name = value
```

## Example 1 — Creating Variables

```
# Creating variables
name = "Rohit"      # String variable
age = 20             # Integer variable
height = 5.8         # Float variable (decimal)
is_student = True    # Boolean variable (True/False)

# Displaying them
print(name)          # Output: Rohit
print(age)            # Output: 20
print(height)         # Output: 5.8
print(is_student)     # Output: True
```

## Key Points About Variables

- Variable names should be meaningful (use `user_age` not just `x`)
- Names are case-sensitive (Name  $\neq$  name)
- Start with a letter or underscore, not a number
- Use underscores for spaces: `my_variable` (not `myVariable`)
- Python doesn't need to declare data type (automatic!)

# 2 DATA TYPES — Different Kinds of Data

Python has 4 main data types:

| Data Type | Example          | Description                   |
|-----------|------------------|-------------------------------|
| int       | age = 25         | Whole numbers (no decimals)   |
| float     | height = 5.8     | Decimal numbers               |
| str       | name = "Rohit"   | Text / characters (in quotes) |
| bool      | is_active = True | True or False only            |

## Example — Using Different Data Types

```
# Different data types
age = 25                # int    - whole number
price = 99.99           # float  - decimal
name = "Rohit Kumar"    # str    - text (use quotes!)
is_premium = True       # bool   - True/False

# Check data type with type()
print(type(age))        # Output: <class 'int'>
print(type(price))      # Output: <class 'float'>
print(type(name))       # Output: <class 'str'>
print(type(is_premium)) # Output: <class 'bool'>
```

**Important: Strings must be in quotes!**

✓ Correct: name = "John" or name = 'John'

✗ Wrong: name = John (Python will think John is a variable)

## 3 input() & print() — Communication with User

### print() — Displaying Output

The print() function displays values on the screen.

#### Example 1 — Simple Print

```
# Basic print
print("Hello, World!")      # Output: Hello, World!

# Print variables
name = "Rohit"
age = 20
print(name)
print(age)
# Output:
# Rohit
# 20

# Print multiple values
print("Name:", name, "Age:", age)
# Output: Name: Rohit Age: 20
```

### input() — Getting User Data

The input() function asks the user to type something and stores it as text.

#### Example 2 — Taking User Input

```
# Get user input
name = input("Enter your name: ")    # User types: Rohit
# name now contains: 'Rohit'

age = input("Enter your age: ")      # User types: 20
# age now contains: '20' (as TEXT, not a number!)

print("Hello " + name + "! You are " + age + " years old")
# Output: Hello Rohit! You are 20 years old
```

#### Important: input() returns TEXT (string)

Even if the user types a number, input() gives you text.

To use it as a number, convert it:

```
# Convert input to number
age_text = input("Enter age: ")      # User types: 20
age_number = int(age_text)           # Convert to integer
# Now age_number = 20 (as number, not text)

# Short way:
age = int(input("Enter age: "))
```

## 4 OPERATORS — Performing Operations

### A. Arithmetic Operators (Math Operations)

| Operator | Name                | Example | Result |
|----------|---------------------|---------|--------|
| +        | Addition            | 5 + 3   | 8      |
| -        | Subtraction         | 5 - 3   | 2      |
| *        | Multiplication      | 5 * 3   | 15     |
| /        | Division            | 6 / 2   | 3.0    |
| //       | Floor Division      | 7 // 2  | 3      |
| %        | Modulus (Remainder) | 7 % 2   | 1      |
| **       | Exponent (Power)    | 2 ** 3  | 8      |

### Example — Arithmetic Operations

```
a = 10
b = 3
print(a + b)    # 13
print(a - b)    # 7
print(a * b)    # 30
print(a / b)    # 3.333...
print(a // b)   # 3 (floor division - removes decimal)
print(a % b)    # 1 (remainder)
print(a ** 2)   # 100 (a squared)
```

## B. Comparison Operators (Comparing Values)

| Operator | Meaning          | Example | Result |
|----------|------------------|---------|--------|
| ==       | Equal to         | 5 == 5  | True   |
| !=       | Not equal to     | 5 != 3  | True   |
| >        | Greater than     | 5 > 3   | True   |
| <        | Less than        | 5 < 3   | False  |
| >=       | Greater or equal | 5 >= 5  | True   |
| <=       | Less or equal    | 5 <= 3  | False  |

### Example — Comparison Operations

```
x = 10
y = 20
print(x == y)    # False (10 is not equal to 20)
print(x != y)    # True  (10 is not equal to 20)
print(x > y)     # False (10 is not greater than 20)
print(x < y)     # True  (10 is less than 20)
print(x >= 10)   # True  (10 is equal to 10)
print(y <= 20)   # True  (20 is equal to 20)
```

## C. Logical Operators (and, or, not)

| Operator | Meaning           | Example        | Result |
|----------|-------------------|----------------|--------|
| and      | Both must be True | True and False | False  |
| or       | At least one True | True or False  | True   |
| not      | Opposite          | not True       | False  |

### Example — Logical Operations

```
age = 20
income = 50000

# and - both must be True
print(age > 18 and income > 30000)  # True and True = True

# or - at least one must be True
print(age < 18 or income > 30000)   # False or True = True

# not - opposite
print(not (age < 18))                # not False = True
```

## 5 PRACTICAL EXAMPLE — Simple Calculator

**Task:** Build a calculator that performs all basic operations on two numbers.

### Calculator — Step by Step

```
# Simple Calculator - Step by Step

# Step 1: Get input from user
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# Step 2: Perform calculations
addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2
division = num1 / num2
remainder = num1 % num2

# Step 3: Display results
print("\n==== CALCULATOR RESULTS =====")
print(f"{num1} + {num2} = {addition}")
print(f"{num1} - {num2} = {subtraction}")
print(f"{num1} * {num2} = {multiplication}")
print(f"{num1} / {num2} = {division}")
print(f"{num1} % {num2} = {remainder}")
print("=====")
```

### Sample Output

```
Enter first number: 10
Enter second number: 3

==== CALCULATOR RESULTS =====
10.0 + 3.0 = 13.0
10.0 - 3.0 = 7.0
10.0 * 3.0 = 30.0
10.0 / 3.0 = 3.333...
10.0 % 3.0 = 1.0
=====
```

### Breaking Down the Calculator

- `float(input())`: gets user input and converts it to a decimal number
- Arithmetic operations: performs `+`, `-`, `*`, `/`, `%`
- f-strings: `f"{num1} + {num2} = {addition}"` — puts variables inside text
- `print()`: displays the formatted result

## 6 COMMON MISTAKES TO AVOID

### 1. Forgetting Quotes for Strings

|           |                            |
|-----------|----------------------------|
| ✗ Wrong   | <code>name = John</code>   |
| ✓ Correct | <code>name = "John"</code> |

### 2. Using = for Comparison

|           |                            |
|-----------|----------------------------|
| ✗ Wrong   | <code>if age = 20:</code>  |
| ✓ Correct | <code>if age == 20:</code> |

### 3. Forgetting to Convert Input

|           |                                                     |
|-----------|-----------------------------------------------------|
| ✗ Wrong   | <code>age = input("Age: ") then age + 1</code>      |
| ✓ Correct | <code>age = int(input("Age: ")) then age + 1</code> |

### 4. Variable Names with Spaces

|           |                                |
|-----------|--------------------------------|
| ✗ Wrong   | <code>my name = "Rohit"</code> |
| ✓ Correct | <code>my_name = "Rohit"</code> |

### 5. Case Sensitivity

|           |                                                   |
|-----------|---------------------------------------------------|
| ✗ Wrong   | Name and name are different variables             |
| ✓ Correct | Use consistent naming: name (lowercase preferred) |

## 7 QUICK REFERENCE GUIDE

| Concept          | Syntax                           | Example                                      |
|------------------|----------------------------------|----------------------------------------------|
| Variable         | <code>name = value</code>        | <code>age = 20</code>                        |
| Print            | <code>print(value)</code>        | <code>print(age)</code>                      |
| Input            | <code>var = input(prompt)</code> | <code>name = input("Enter: ")</code>         |
| Convert to Int   | <code>int(value)</code>          | <code>age = int(input("Age: "))</code>       |
| Convert to Float | <code>float(value)</code>        | <code>price = float(input("Price: "))</code> |
| Check Type       | <code>type(value)</code>         | <code>type(age)</code>                       |
| Add Numbers      | <code>x + y</code>               | <code>10 + 5 = 15</code>                     |
| Compare          | <code>x == y</code>              | <code>10 == 10 = True</code>                 |

## Key Takeaways

- ✓ Variables store data — think of them as labeled boxes
- ✓ Data types: int (numbers), float (decimals), str (text), bool (True/False)
- ✓ input() gets data from the user (always as text initially)
- ✓ print() displays output on screen
- ✓ Convert input using int() or float() to change text into numbers
- ✓ Arithmetic: +, -, \*, /, //, %, \*\*
- ✓ Comparison: ==, !=, >, <, >=, <=
- ✓ Logical: and, or, not

## Next Steps

- Practice: try the 20 practice questions for Day 1
- Build: create a complete calculator (add, subtract, multiply, divide)
- Experiment: write small programs using variables and operators
- Upload: save your calculator code on GitHub